

NeuroShell: Dengue Outbreak Prediction Using Machine Learning and Weather Data

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

In Sri Lanka, dengue fever is a serious public health issue, especially in the Colombo District, which has the highest dengue case incidence in the country. The majority of current dengue surveillance techniques rely on retrospective trend analysis and descriptive statistics, which are inadequate for producing proactive early warnings. The implementation of machine learning-based temporal prediction models that can anticipate dengue epidemic risk levels with enough accuracy and lead time for useful public health applications is severely lacking.

Using historical dengue surveillance data and meteorological variables for the Colombo District, this project focuses on creating, assessing, and refining a suite of machine learning and deep learning models that can forecast dengue outbreak risk levels and outbreak probabilities beforehand. Dengue outbreak prediction is formulated by the suggested system as a probabilistic forecasting challenge to estimate epidemic likelihood over one to four week prediction horizons as well as a time-based classification problem (Low, Medium, High risk).

The approach combines deep learning time-series models like Long Short-Term Memory (LSTM) networks with traditional machine learning methods like Random Forest and XGBoost. To stop data leakage and guarantee accurate forecasting performance, time-series-aware validation techniques including rolling window methods and walk-forward validation are used. Grid search and random search are used for hyperparameter tuning, and models are assessed using accuracy, F1-score, ROC-AUC, and RMSE metrics. This component is unusual since it uses localised Sri Lankan data to systematically compare deep learning and traditional temporal models, together with probabilistic risk forecasting and multi-horizon evaluation.

- Keywords: Keywords: Dengue prediction, machine learning, LSTM, XGBoost, Random Forest, time-series forecasting, Sri Lanka, outbreak risk, probabilistic forecasting.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ML	Machine Learning
DL	Deep Learning
LSTM	Long Short-Term Memory
RF	Random Forest
XGBoost	Extreme Gradient Boosting
NDCU	National Dengue Control Unit
EDA	Exploratory Data Analysis
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
RMSE	Root Mean Squared Error
F1	F1-Score (Harmonic Mean of Precision and Recall)
SDG	Sustainable Development Goal
AI	Artificial Intelligence
API	Application Programming Interface
MOH	Medical Officer of Health
WHO	World Health Organization

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1. INTRODUCTION

1.1 Background and Context

One of the biggest public health issues facing tropical and subtropical nations, including Sri Lanka, is dengue fever, a mosquito-borne viral disease mostly spread by the *Aedes aegypti* mosquito [1]. Rainfall, temperature, and humidity are the main factors influencing mosquito breeding cycles and the dynamics of illness transmission, making the disease extremely climate-sensitive [2]. Due to its dense urban population, high mobility, and climate that encourages mosquito growth, the Colombo District typically records the highest incidence of dengue cases nationwide during Sri Lanka's biannual dengue outbreaks [3].

The National Dengue Control Unit (NDCU) of Sri Lanka reports that dengue cases have been steadily increasing over the last ten years, with notable outbreak peaks in 2017 and 2019, both of which had more than 100,000 reported cases nationwide [3]. Even though historical dengue surveillance data and weather records are widely accessible, Sri Lanka's current dengue control strategy is still primarily reactive. The effectiveness of early intervention efforts is limited since preventive actions and the deployment of health resources are usually only started after a significant rise in reported cases has already occurred [3].

Dengue transmission dynamics are intrinsically complicated, comprising lagged and non-linear connections between environmental factors and mosquito population dynamics, which in turn influence case incidence with temporal delays corresponding to mosquito development and incubation periods [2]. Due to their inability to simulate high-dimensional, non-linear temporal correlations, conventional statistical techniques like ARIMA models and linear regression are unable to sufficiently capture these dynamics [4].

Table 1: Functional Components and Novel Research Contributions of NeuroShell

Components	Research Tasks	Novelty / Research Contribution
Exploratory Data Analysis & Risk Formulation	Temporal trend analysis, seasonality decomposition, autocorrelation study, outbreak cycle detection, risk threshold definition.	First systematic EDA of Colombo District dengue time-series data with biologically grounded risk level threshold derivation aligned to NDCU epidemiological benchmarks.
Time-Lag Feature Engineering	Generation of 1–4 week lag features for all meteorological variables, rolling window aggregate statistics, seasonal and temporal indicators	Biologically motivated lag feature construction grounded in <i>Aedes aegypti</i> mosquito development cycle dynamics, applied specifically to localized Sri Lankan climate and surveillance data
Time-Series-Aware Validation	Walk-forward validation, rolling window splits, multi-horizon evaluation across 1–4 week horizons	Rigorous temporal validation methodology preventing future data leakage, producing realistic and reproducible forecasting performance estimates for Sri Lankan dengue data

Probabilistic Outbreak Forecasting	Outbreak probability score generation, Brier score and log-loss evaluation, calibration curve analysis	Novel formulation of dengue prediction as a probabilistic forecasting task producing calibrated outbreak likelihood estimates, extending beyond conventional case count regression
Model Selection & Integration	Ensemble construction, multi-horizon stability analysis, FastAPI endpoint development, model serialization	End-to-end deployment-ready ML pipeline exposing risk level and probability outputs via REST API for seamless integration with dashboard and NLP system components

1.2 Literature Review

Numerous studies have looked into the use of machine learning and statistical techniques to predict dengue incidence. To model temporal patterns in dengue case data, early research primarily used time-series statistical techniques such as linear regression and Seasonal Autoregressive Integrated Moving Average (SARIMA) models [4]. The ability of these approaches to capture the non-linear and lagged relationships between climatic variables and dengue transmission dynamics is fundamentally limited, despite their interpretability and computational efficiency [4].

Later studies focused on machine learning techniques that could model non-linear dependencies. Using meteorological variables as primary predictors, Xu et al. [5] showed that ensemble techniques such as Random Forest and Gradient Boosting consistently outperformed conventional statistical models in predicting dengue case counts across multiple tropical regions. The significance of feature engineering was emphasized by their work, especially the incorporation of lagged climatic variables that correspond to mosquito development cycles of one to four weeks.

Because deep learning techniques, especially LSTM networks, can represent long-term temporal connections in sequential data, they are being used more and more to forecast dengue [6]. In situations where the prediction signal is dispersed over long temporal periods, like seasonal dengue transmission cycles, LSTM models have proven to perform better than conventional machine learning techniques. However, implementation in resource-constrained environments is practically challenging because to the computational needs of LSTM training and the need for significant historical data.

In a spatiotemporal investigation of dengue incidence in Sri Lanka, Perera and Fernando [7] found a considerable geographical clustering of dengue risk in the Colombo District and showed the use of district-level meteorological data for predictive modelling. In contrast to national-level aggregate techniques, which may obscure significant sub-regional heterogeneity in dengue risk, their findings highlight the significance of localized, spatially tailored models.

The emphasis on point prediction of case counts rather than risk-level categorization or probabilistic forecasting is a prevalent drawback throughout current studies. For public health decision-makers who must allocate resources and issue warnings based on categorical risk assessments rather than exact number case estimates, risk-level categorization results are more immediately actionable [5]. Additionally, the majority of current models do not provide calibrated probability estimates for the occurrence of outbreaks, which restricts their use in frameworks for uncertainty-aware decision-making.

Another significant drawback of the current research is the time-series validation technique. Overly optimistic performance estimates that do not accurately represent real-world predicting capacity result from the use of random train-test splits in many research, which enable future data to impact prior forecasts [8]. When assessing time-series forecasting models in practical operational environments, walk-forward validation and rolling window techniques are regarded as recommended practices.

1.3 Research Gap

This research is motivated by a number of crucial gaps in the expanding body of literature on machine learning applications in dengue prediction. First, most current research concentrates on national or large regional datasets, which restricts their ability to produce the district-level, localized projections needed for focused public health interventions in Sri Lanka [5][7]. Second, the majority of current models are assessed in offline experimental environments using random train-test splits, which produce inflated performance estimates and fail to accurately capture the temporal nature of the prediction issue [8].

Third, comprehensive comparison studies that use localized dengue data from Sri Lanka to assess both deep learning time-series models and traditional machine learning models under uniform assessment settings are lacking. Fourth, despite being more directly related to public health decision-making, the formulation of dengue prediction as a risk-level classification task and probabilistic forecasting problem is rarely investigated. Instead, the majority of current studies formulate dengue prediction as a regression problem targeting case count estimation.

This research context is poised to bridge this gap by merging historical data on tea grade quantities with data-driven inputs, thereby formulating predictive models that resonate with the institute's commitment to innovation. By interweaving variables encompassing historical trends, environmental nuances, and evolving consumer preferences, these models hold the promise of facilitating informed and proactive decision-making. Such a proactive stance aligns harmoniously with the ethos of the Tea Research Institute of Sri Lanka, positioning it at the vanguard of embracing cutting-edge technologies to further enhance the tea industry.

Using time-series-aware validation techniques, risk-level classification outputs, and probabilistic forecasting capabilities across multiple prediction horizons, this study develops a rigorous, locally grounded comparative study of classical machine learning and deep learning temporal models for dengue outbreak risk prediction in the Colombo District.

Comparison of Related Studies

Table 2:Session Management Limitations in Existing Cybersecurity Automation Frameworks

Study	Method Used	Dataset / Region	Prediction Type	Limitation
Naish et al. (2014)	SARIMA, linear regression	Multi-country tropical data	Case count estimation	Cannot capture non-linear relationships
Xu et al. (2020)	Random Forest, Gradient Boosting	Multiple tropical regions	Case count estimation	Random train-test split used; no temporal validation
Hochreiter & Schmidhuber (1997)	LSTM neural network	General sequential data	Sequence modelling	High computational demand; needs large dataset

Perera & Fernando (2021)	Spatiotemporal analysis	Sri Lanka (district level)	Spatial risk mapping	No ML-based temporal prediction component
Rocklöv & Johansson (2019)	Statistical climate modelling	Global	Transmission risk estimation	No localized or district-level prediction
Hyndman & Athanasopoulos (2021)	Time-series forecasting methods	General	Case count forecasting	No risk-level classification or probabilistic output

2.OBJECTIVE

2.1 Main Objective

To design, develop, evaluate, and optimize a suite of classical machine learning and deep learning time-series models — including Random Forest, XGBoost, and LSTM networks — capable of predicting dengue outbreak risk levels (Low, Medium, High) and estimating outbreak probabilities across 1 to 4 week forecasting horizons for the Colombo District of Sri Lanka, by leveraging historical dengue surveillance data from the National Dengue Control Unit and meteorological variables from the Department of Meteorology, employing time-series-aware validation methodologies to ensure realistic forecasting performance, with the ultimate aim of generating accurate, localized, and interpretable early warnings that support proactive dengue prevention and data-driven public health decision-making.

2.2 Specific Objectives

- Dengue Outbreak Risk Formulation and Exploratory Analysis.**
 - This Formulate dengue outbreak prediction as both a time-based classification problem (Low, Medium, High risk) and a probabilistic forecasting task to estimate outbreak likelihood.
 - Perform exploratory data analysis on temporal dengue case patterns to identify seasonality, outbreak cycles, trend components, and anomalies relevant to model design.
 - Define and validate risk level thresholds for the Colombo District based on historical case distributions and NDCU epidemiological benchmarks.
- Machine Learning Model Development and Training.**
 - Train and evaluate classical machine learning models including Random Forest and XGBoost on the processed dengue and meteorological dataset.
 - Develop LSTM deep learning models to capture non-linear long-term temporal dependencies between weather variables and dengue case incidence.
 - Generate biologically motivated time-lag features (1 to 4 weeks) based on mosquito lifecycle dynamics to serve as input features for all models.
- Time-Series-Aware Validation and Performance Evaluation.**
 - Apply time-series-aware data splitting strategies including walk-forward validation and rolling window approaches to prevent future data leakage.

- Evaluate model performance across multiple forecasting horizons (1, 2, 3, and 4 weeks ahead) using accuracy, F1-score, ROC-AUC, Precision, Recall, and RMSE.
- Conduct calibration analysis of probabilistic forecasts to assess the reliability of outbreak probability estimates.

4. Hyperparameter Optimization and Model Selection.

- Optimize model performance using hyperparameter tuning techniques including grid search and random search across all candidate models.
- Analyze model stability across forecasting horizons and identify the most reliable individual model or ensemble configuration for operational deployment.
- Document and report the final selected model with full reproducibility details for integration with the broader dengue prediction system.

3.METHODOLOGY

Our Using a Design Science Research methodology, this study focuses on the methodical development, assessment, and optimization of predictive artefacts in the form of machine learning models for dengue epidemic risk forecasting. Data preparation, model creation, validation, optimization, and integration are all covered by the methodology's organized five-phase workflow.

3.1 Research Approach

This research adopts a Design Science Research methodology, focusing on the systematic construction, evaluation, and optimization of machine learning artefacts for dengue outbreak risk forecasting. An experimental comparative approach is employed to rigorously isolate and evaluate the temporal prediction problem, determining the most effective model architecture for localized dengue forecasting in the Colombo District under consistent and realistic experimental conditions..

The research proceeds through four iterative phases:

- I. **EDA & Problem Formulation:** Exploratory analysis of historical dengue and meteorological data to identify seasonal patterns, outbreak cycles, and autocorrelation structure, followed by risk level threshold definition.
- II. **Feature Engineering & Data Preparation:** Construction of biologically motivated time-lag features, rolling window statistics, seasonal indicators, and normalization of all variables for model-specific input requirements.
- III. **Model Development & Training:** Parallel training of Random Forest, XGBoost, and LSTM models on the processed dataset, followed by ensemble construction combining outputs from all three tracks.
- IV. **Validation, Tuning & Selection:** Time-series-aware walk-forward validation across 1–4 week horizons, hyperparameter optimization via grid and random search, and final model selection based on multi-horizon stability and performance metrics.

3.2 Technical Methods and Algorithms

3.2.1 Time-Lag Feature Engineering

The core feature engineering challenge is the lagged relationship between meteorological variables and dengue case incidence. The *Aedes aegypti* mosquito development cycle spans one to four weeks, meaning that rainfall and temperature conditions at time t influence dengue case counts at time $t+1$ to $t+4$. A biologically motivated lag feature pipeline generates lagged copies of all meteorological variables at one, two, three, and four week intervals. Rolling window aggregate statistics including mean, variance, and maximum are computed over two to six week windows to capture medium-term weather trends. Additional temporal features including week of year, month, and binary seasonal indicators are incorporated to encode known seasonal transmission patterns..

3.2.2 LSTM Deep Learning Model Development

A stacked Long Short-Term Memory network is implemented using TensorFlow and Keras to capture non-linear long-term temporal dependencies in the sequential dengue and meteorological data. The architecture comprises two stacked LSTM layers with 128 and 64 hidden units respectively, each with dropout regularization at a rate of 0.2, followed by a dense layer with ReLU activation and an output layer with Softmax activation for risk level classification and Sigmoid activation for probabilistic outbreak forecasting. Input sequences span eight weeks of feature-engineered data to provide sufficient temporal context for the model.

3.2.3 Classical Machine Learning Model Development

Random Forest and XGBoost classifiers are trained on the tabular feature-engineered dataset using scikit-learn. Random Forest leverages bootstrap aggregation across an ensemble of decision trees to produce robust risk level classifications and generates feature importance scores that support model interpretability. XGBoost applies sequential gradient boosting with L1 and L2 regularization to prevent overfitting on the limited historical dataset. Both models produce categorical risk level outputs (Low, Medium, High) and class probability estimates used as inputs to the ensemble layer.

3.2.4 Ensemble Layer and Prediction Output

soft voting ensemble layer combines the class probability outputs of the Random Forest, XGBoost, and LSTM models to produce a final risk level prediction and outbreak probability score. The ensemble is evaluated against individual model tracks to determine whether combination improves multi-horizon stability. Final outputs include a categorical risk level label, an outbreak probability score between 0.0 and 1.0, and predictions across one to four week forecasting horizons, all serialized and exposed via a FastAPI REST endpoint for consumption by the dashboard and NLP components.

3.3 Tools and Platforms with Justification

Machine Learning Framework (scikit-learn): scikit-learn is selected for Random Forest and XGBoost model training due to its mature, well-documented API, native support for GridSearchCV and RandomizedSearchCV hyperparameter optimization, and comprehensive suite of classification and regression evaluation metrics directly applicable to the dengue prediction task.

Deep Learning Framework (TensorFlow and Keras): TensorFlow with the Keras high-level API is selected for LSTM model development due to its flexible sequential model construction, native GPU acceleration support via Google Colab, and TensorFlow SavedModel serialization format enabling reproducible deployment. Keras's callback system supports early stopping and learning rate scheduling during LSTM training.

Data Processing (Pandas and NumPy): Pandas provides the primary data manipulation layer for time-series alignment, lag feature generation, and rolling window computation. NumPy supports efficient numerical operations on the feature matrix. Both libraries integrate seamlessly with scikit-learn and TensorFlow input pipelines.

Compute Environment (Google Colab with GPU): Google Colab with GPU runtime is selected for LSTM model training to provide the necessary computational acceleration without requiring dedicated on-premise hardware, reducing LSTM training time from hours to minutes for the dataset scale of this project.

Backend API (FastAPI): FastAPI provides the REST endpoint interface exposing model predictions to the dashboard visualization and NLP components. FastAPI's asynchronous architecture enables concurrent handling of prediction requests and its automatic OpenAPI documentation simplifies integration with other system components.

Version Control (Git and GitHub): Git and GitHub are used for version control of all model training scripts, feature engineering pipelines, and evaluation notebooks, ensuring reproducibility and enabling collaborative development across the project team.

3.4 Data Collection and Ethical Considerations

Dataset Construction: The primary dataset consists of weekly dengue case counts for the Colombo District obtained from the National Dengue Control Unit of the Ministry of Health, Sri Lanka, under formal written data access authorization. Meteorological data including weekly rainfall, maximum and minimum temperature, relative humidity, and wind speed is obtained from the Department of Meteorology of Sri Lanka under a standard institutional data sharing agreement. The combined dataset spans the period 2015 to 2024, providing a sufficient historical window to capture multiple seasonal dengue transmission cycles and outbreak peaks.

Ethical Considerations: All dengue case data is aggregated at the district level and contains no personally identifiable patient information, ensuring full compliance with patient privacy requirements. All data is used exclusively for academic research purposes under the terms of the respective data sharing agreements. The research protocol is submitted for ethical review under SLIIT institutional guidelines prior to data collection. No human participants are recruited or involved in any data collection activities, and no real-world clinical or personal health data beyond aggregated surveillance records is accessed.

3.5 Validation Strategy and Performance Metrics

Risk Level Classification Accuracy: The accuracy with which the trained models correctly classify weekly dengue risk levels as Low, Medium, or High, measured using weighted F1-score and ROC-AUC against the held-out test set produced by walk-forward validation. Target: minimum weighted F1-score of 0.75 and ROC-AUC of 0.80 across all prediction horizons.

Probabilistic Forecasting Calibration: The degree to which the model's outbreak probability scores accurately reflect true empirical outbreak frequencies, measured using Brier Score and calibration curve analysis. Target: Brier

Score below 0.15, indicating well-calibrated probability estimates suitable for uncertainty-aware public health decision-making.

Multi-Horizon Forecast Stability: The consistency of model performance across one, two, three, and four week prediction horizons, assessed by comparing weighted F1-scores across all horizons. Target: performance degradation of no more than 10% between the one week and four week horizons, ensuring operational utility at extended forecast lead times.

Hyperparameter Optimization Improvement: The relative improvement in weighted F1-score achieved by the tuned model compared to the baseline untuned model, measured on the same walk-forward validation splits. Target: minimum 5% improvement in weighted F1-score following grid and random search optimization.

Inference Latency: The time required to generate a single weekly risk level prediction and outbreak probability score from preprocessed input features. Target: under 2 seconds for all model configurations to ensure compatibility with near-real-time dashboard update requirements.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The ML Engineer component is designed as a modular, phase-driven machine learning pipeline that serves as the core temporal prediction intelligence layer of the Dengue Outbreak Prediction System. The component functions as the primary analytical engine that transforms preprocessed dengue surveillance and meteorological data into actionable outbreak risk predictions and probability estimates for the Colombo District. The architecture follows a sequential pipeline design in which each phase is responsible for a clearly defined stage of the modelling workflow, from exploratory analysis and feature engineering through model training, validation, and deployment. The entire pipeline is coordinated through a FastAPI-based REST API that exposes prediction outputs to the dashboard visualization and NLP components of the broader integrated system.

Component	Name	Description
Phase 1	EDA & Problem Formulation	Conducts exploratory analysis of historical dengue and meteorological data to identify seasonal patterns, outbreak cycles, and autocorrelation structure, and derives empirically grounded risk level thresholds.
Phase 2	Feature Engineering Pipeline	Constructs biologically motivated time-lag features spanning one to four weeks, rolling window aggregate statistics, and seasonal indicators to produce ML-ready input datasets.

Phase 3	Model Training Tracks	Trains three parallel model tracks — Random Forest, XGBoost, and LSTM — on the feature-engineered dataset and combines their outputs through a soft voting ensemble layer.
Phase 4	Validation & Evaluation Engine	Applies walk-forward and rolling window validation across one to four week prediction horizons, evaluating models using classification, regression, and probabilistic calibration metrics.
Phase 5	Prediction API	Exposes finalized model predictions — categorical risk level and outbreak probability score — via a FastAPI REST endpoint for consumption by dashboard and NLP components.

Table 4 - Core Components of the NeuroShell Framework

[Figure 1 - High-Level System Architecture of the NeuroShell Framework -- to be inserted here]

[Figure 2 - Complete System Architecture of the NeuroShell Framework -- to be inserted here]

4.2 ML Engineer Component — Internal Architecture

The ML Engineer component represents the primary research focus of this project. It functions as the temporal prediction layer of the Dengue Outbreak Prediction System, transforming preprocessed surveillance and meteorological data into calibrated, multi-horizon dengue outbreak risk assessments for the Colombo District. The component's internal architecture implements five tightly integrated sub-modules.

4.2.1 EDA & Risk Formulation Module the EDA module conducts comprehensive

exploratory analysis of the historical dengue case and meteorological dataset to characterize the temporal structure of dengue transmission in the Colombo District. Seasonal decomposition, autocorrelation analysis, and outbreak cycle detection are performed to identify patterns relevant to model design. Risk level thresholds defining Low, Medium, and High risk categories are derived from the empirical distribution of weekly dengue case counts and validated against NDCU epidemiological benchmarks.

4.2.2 Feature Engineering Pipeline

The feature engineering pipeline constructs the complete set of model input features from the preprocessed dataset. Biologically motivated time-lag features are generated for all meteorological variables at one, two, three, and four week intervals, reflecting the development timeline of the *Aedes aegypti* mosquito. Rolling window aggregate statistics including mean, variance, and maximum are computed over two to six week windows. Temporal and seasonal indicator features are appended, and all features are normalized using min-max or z-score normalization as appropriate for each model type.

4.2.3 Model Training Tracks Three parallel model tracks are trained on the feature-engineered dataset. The Random Forest track trains an ensemble classifier using scikit-learn, leveraging bootstrap aggregation and feature importance estimation. The XGBoost track applies gradient boosting with L1 and L2 regularization. The LSTM track implements a stacked recurrent neural network using TensorFlow and Keras, with two LSTM layers of 128 and 64 hidden units respectively, dropout regularization at 0.2, and output layers for both classification and probabilistic forecasting. A soft voting ensemble layer combines class probability outputs from all three tracks to produce the final prediction.

4.2.4 Validation & Evaluation Engine The validation engine applies walk-forward validation across the full historical dataset, training on all data up to time *t* and evaluating on the subsequent *k* weeks, repeated across multiple temporal splits. Rolling window validation additionally assesses model stability over time. Models are evaluated across one, two, three, and four week prediction horizons using weighted F1-score, ROC-AUC, RMSE, Brier Score, and calibration curve analysis.

4.2.5 Prediction API The finalized model is serialized using joblib and TensorFlow SavedModel and exposed via a FastAPI REST endpoint. The API receives preprocessed weekly feature inputs, generates a categorical risk level prediction and an outbreak probability score for each of the one to four week horizons, and returns structured JSON responses for consumption by the dashboard and NLP components.

4.3 Component Interactions and Data Flow

Step	Process Stage	Description	Example Output
1	Data Ingestion	Preprocessed weekly dengue case counts and meteorological variables are received from the Data Engineering component.	Weekly feature matrix for Colombo District
2	Feature Engineering	Time-lag features, rolling window statistics, and seasonal indicators are generated from the input data.	Feature-engineered dataset with 30+ input variables
3	Model Inference	All three model tracks generate class probability outputs from the feature-engineered input.	RF: [0.1, 0.3, 0.6], XGB: [0.08, 0.28, 0.64], LSTM: [0.12, 0.31, 0.57]
4	Ensemble Aggregation	Soft voting ensemble layer averages class probabilities across all three tracks to produce the final risk level and outbreak probability.	Risk: High (0.62 probability)

5	Multi-Horizon Forecasting	Final predictions are generated for one, two, three, and four week ahead horizons.	Week 1: High, Week 2: High, Week 3: Medium, Week 4: Medium
6	API Response	Structured prediction output is serialized to JSON and returned via the FastAPI endpoint to the dashboard and NLP components.	{"risk":"High", "probability":0.62, "horizon":"1w"}

Table 7 - NeuroShell System Execution Workflow

4.4 Scalability and Security Considerations

Scalability: The modular pipeline architecture enables extension to additional districts beyond Colombo with minimal architectural changes — new district datasets are processed through the same feature engineering and model training pipeline with district-specific parameterization. Model retraining pipelines are designed to accommodate periodic updates as new surveillance data becomes available, ensuring predictions remain calibrated to current epidemiological conditions. The FastAPI endpoint architecture supports horizontal scaling to handle concurrent prediction requests from multiple system components.

Security: The prediction API implements token-based authentication to restrict access to authorized system components. All input features are validated at the API layer before model inference to prevent malformed data from affecting prediction outputs. Model artifacts are stored with version control to ensure reproducibility and enable rollback to prior model versions if performance degradation is detected following retraining.

5.USER REQUIREMENTS

5.1 Stakeholder identification

1. Primary stakeholders: Public health officers at the NDCU, Medical Officers of Health (MOH) at district level, and epidemiologists responsible for dengue surveillance in the Colombo District.
2. Secondary stakeholders: System developers and data engineers integrating the ML component with the broader system, and academic supervisors validating model performance.

5.2 Functional requirements

1. Prediction:
 - System shall predict dengue risk level (Low / Medium / High) for the Colombo District on a weekly basis.
 - The system will produce outbreak probability estimates for prediction horizons of one, two, three, and four weeks.
2. Integration:
 - Model predictions will be made available by the system through a REST API endpoint tthe dashboard and NLP components can access.

3. Maintenance:

- When new surveillance data becomes available, the system will allow for periodic model retraining.

4. Logging:

- System shall log all predictions with timestamps and input feature values for auditability.

5.3 Non-functional requirements

1. Performance:

- a. Model inference for a single weekly prediction shall complete within 2 seconds.

2. Accuracy:

- a. Final ensemble model shall achieve a minimum weighted F1-score of 0.75 on the held-out test set.

3. Reliability:

- a. The prediction API shall maintain 99% uptime during operational deployment periods.

4. Security:

- a. API endpoints shall require token-based authentication; all data transmissions shall use HTTPS.

5. Scalability:

- a. System shall accommodate additional districts beyond Colombo with minimal architectural changes.

6.COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market is the National Dengue Control Unit (NDCU) and district-level Medical Officers of Health (MOH) within the Sri Lanka Ministry of Health, who require localized, data-driven early warning tools to support proactive dengue prevention strategies. The secondary target market comprises municipal health authorities and public health agencies in other dengue-endemic countries across South and Southeast Asia, including India, Bangladesh, Thailand, and Malaysia, seeking adaptable and locally calibrated outbreak prediction frameworks.

Value Proposition

The ML Engineer component delivers one to four week advance warning of dengue outbreak risk levels and calibrated outbreak probability estimates, enabling health authorities to proactively deploy fumigation resources, issue timely public advisories, and prepare hospital capacity before outbreaks escalate. Unlike existing surveillance tools that rely on retrospective case count reporting, this system provides forward-looking, interpretable risk classifications directly actionable by non-technical public health officials.

Revenue Model

The system is commercialized through three complementary revenue streams. A government licensing model provides annual licensing agreements with the Ministry of Health for deployment of the integrated dengue prediction platform. A SaaS subscription model offers cloud-hosted access to the prediction API and dashboard for regional health authorities in dengue-endemic countries. A research partnership model generates revenue through collaboration agreements with academic institutions and international health organizations including WHO for model validation, expansion, and co-publication.

Cost Estimation

Initial development costs are primarily absorbed within the academic research project. Post-development deployment costs include cloud server hosting for the FastAPI prediction endpoint estimated at approximately Rs. 15,000 per month, periodic model retraining compute costs estimated at Rs. 5,000 per retraining cycle, and ongoing data acquisition and maintenance costs estimated at Rs. 10,000 annually. Total first-year operational cost is estimated at Rs. 195,000.

Pricing Strategy

Government licensing is priced as an annual flat-fee arrangement negotiated directly with the Ministry of Health, estimated at USD 5,000 to USD 15,000 per year depending on deployment scale and support requirements. SaaS subscriptions for regional health authorities are priced on a tiered model based on the number of districts covered and prediction frequency, starting at USD 200 per month. Research partnership agreements are structured as cost-recovery contracts covering compute, data, and researcher time.

Competitive Advantage

The primary competitive advantage lies in the localized calibration of all models to Sri Lankan epidemiological and meteorological data, combined with multi-horizon probabilistic forecasting capability not available in existing public health surveillance tools. The system's time-series-aware

validation methodology produces realistic, trustworthy performance estimates that differentiate it from academic models evaluated under artificially optimistic conditions. The integrated framework combining ML prediction, NLP intelligence, and interactive visualization creates a defensible multi-component platform with high institutional switching costs.

Intellectual Property Considerations

The biologically informed time-lag feature engineering methodology, grounded in *Aedes aegypti* mosquito development cycle dynamics and applied to localized Sri Lankan surveillance data, represents a potentially patentable novel data preparation approach for dengue prediction systems. The multi-track ensemble architecture combining Random Forest, XGBoost, and LSTM outputs under walk-forward validation is a candidate for copyright protection as a software artefact. All datasets curated and processed within this research remain subject to the data sharing agreements with the NDCU and the Department of Meteorology, and cannot be commercialized independently without explicit authorization from those institutions.

7.BUDGET AND JUSTIFICATION

Requirement	Cost (Rs.)	Justification	Requirement
Cloud GPU compute for LSTM training (Google Colab Pro subscription)	12,000.00	GPU acceleration required for LSTM model training; Colab Pro provides sufficient GPU hours at low cost.	Cloud GPU compute for LSTM training (Google Colab Pro subscription)
Data collection travel costs (NDCU, Dept. of Meteorology)	10,000.00	Required to collect historical dengue and meteorological datasets from government institutions in person.	Data collection travel costs (NDCU, Dept. of Meteorology)
Model deployment server (cloud hosting for FastAPI)	8,000.00	Hosting the prediction API endpoint on a cloud server (e.g., AWS Free Tier or Heroku) for system integration.	Model deployment server (cloud hosting for FastAPI)
Software and library licenses	0.00	All required ML libraries (scikit-learn, TensorFlow, XGBoost) are open-source and available at no cost.	Software and library licenses
Internet and data access costs	4,000.00	High-speed internet required for data download, model training on cloud, and collaborative development.	Internet and data access costs
Research paper access fees	3,000.00	Access to IEEE, Elsevier, and Springer publications not available through institutional subscriptions.	
Miscellaneous and contingency	3,000.00	Contingency for unexpected expenses including additional cloud compute or	

		storage requirements.	
Total	40,000.00		

8.WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Stage	Tasks	Deliverables
01	Problem Formulation & EDA	Define risk level thresholds Temporal trend analysis Autocorrelation & seasonality study Outbreak cycle identification	EDA report, risk label definitions
02	Feature Engineering	Generate 1-4 week lag features Rolling window statistics Temporal & seasonal indicators Feature normalization	Feature-engineered ML-ready dataset
03	ML Model Development	Random Forest model training XGBoost model training LSTM network architecture design LSTM model training	Trained model artefacts
04	Validation & Evaluation	Walk-forward validation setup Multi-horizon evaluation Calibration analysis Performance benchmarking	Evaluation report with metrics
05	Optimization & Selection	Grid/random search tuning Ensemble model construction Stability analysis across horizons	Optimized, deployment-ready model

		Final model selection	
06	Optimization & Selection	FastAPI endpoint development Model serialization System integration testing Technical documentation	Deployed API, final report

9.GANTT CHART

Task	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Literature Review & Problem Def.												
Data Collection & EDA												
Preprocessing & Feature Engineering												
ML Model Development (RF, XGBoost)												
LSTM Deep Learning Model Dev.												
Time-Series Validation Setup												
Hyperparameter Tuning & Optimization												
Model Evaluation & Comparison												
Probabilistic Forecasting Module												
System Integration & API Dev.												
Testing, Validation & Documentation												
Final Report Submission												

Figure 1: Gantt Chart for the NeuroShell Framework Development Timeline

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10.APPENDICES

Appendix A: Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
NDCU or Meteorology Dept. denies data access	Medium	High	Early formal data request submission; identify alternative open-access datasets as backup
Insufficient historical data for LSTM training	Medium	High	Extend data collection period; apply data augmentation techniques such as sliding window expansion
Model performance below target F1-score of 0.75	Medium	High	Expand feature set; apply additional hyperparameter tuning; explore alternative ensemble configurations
Google Colab GPU quota exceeded during LSTM training	High	Medium	Monitor usage; upgrade to Colab Pro; use Kaggle GPU as alternative compute environment
FastAPI integration failure with dashboard/NLP components	Low	Medium	Define and agree API contract early; implement mock endpoint for parallel development
Team member unavailability due to illness or other reasons	Low	Medium	Maintain shared codebase on GitHub; document all pipeline steps for handover

Appendix B: Proposed LSTM Architecture

- Input Layer — sequence length of 8 weeks, feature dimension equal to number of engineered input features
- LSTM Layer 1 — 128 hidden units, return sequences = True, dropout rate = 0.2
- LSTM Layer 2 — 64 hidden units, return sequences = False, dropout rate = 0.2
- Dense Layer — 32 units, ReLU activation
- Output Layer — Softmax activation for risk level classification (3 classes); Sigmoid activation for probabilistic outbreak forecasting
- Optimizer — Adam, learning rate = 0.001
- Loss — Categorical Cross-Entropy (classification); Binary Cross-Entropy (probabilistic forecasting)

Appendix C: Ethical Clearance Declaration

All dengue case data used in this research is aggregated at the district level and does not contain any personally identifiable patient information. Data will be obtained under formal written authorization from the National Dengue Control Unit and the Department of Meteorology of Sri Lanka under standard institutional data sharing agreements. The research protocol has been prepared for submission under SLIIT ethical review guidelines. No human participants are recruited or involved in any data collection activities. All data is used exclusively for academic research purposes and will not be shared with third parties beyond the project research team and supervisors.

Appendix D: Survey Instruments

A structured questionnaire will be administered to public health officers at the NDCU and district-level Medical Officers of Health to validate the usability and practical relevance of the system's risk level output format and dashboard interface. The questionnaire will cover the following areas:

- Perceived usefulness of 1–4 week advance outbreak risk warnings for resource allocation decisions
- Clarity and interpretability of Low / Medium / High risk level classifications
- Preferred format and frequency of automated dengue risk reports
- Willingness to adopt a ML-based early warning system within existing surveillance workflows
- Suggestions for additional features or data sources relevant to operational dengue control

